Aurora: Module Relay x8

Stakeholders Requirements

Prepared for

Technical Staff

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Disclosure is prohibited

Revision History

Version	Date	Author	Description
0.1.0	2026-07-02	Martin Pellizzer	Starting version

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requirements

Requirement 1

Requirement

The Relay Module shall execute relay-output commands received from CORE

Requirement 2

Requirement

The Relay Module shall provide CORE with output status

Requirement 3

Requirement

The Relay Module shall provide CORE with module diagnostics

Requirement 4

Requirement

The Relay Module shall transition its outputs to safe states when system-level fault

Requirement 5

Requirement

The Relay Module shall provide eight indipendently controlled relay-driven channels

Requirement 6

Requirement

The Relay Module shall manage activation of each channel

Requirement 7

Requirement

The Relay Module shall manage deactivation of each channel

Requirement 8

Requirement

The Relay Module shall monitor output faults

Requirement 9

Requirement

The Relay Module shall communicate channels status to CORE

Requirement 10

Requirement

The Relay Module shall communicate channels diagnostics to CORE

Requirement 11

Requirement

The Relay Module shall communicate to CORE via Modbus RTU protocol

Requirement 12

Requirement

The Relay Module shall communicate to CORE via RS-485 hardware standard

Requirement 13

Requirement

The Relay Module shall enforce local fault-handling

Requirement 14

Requirement

The Relay Module shall turn ON/OFF outputs within specified time

Requirement 15

Requirement

The Relay Module shall prevent one faulty channel to affect other channels

Requirement 16

Requirement

The Relay Module shall maintain the output defined safe behavior under fault conditions

Requirement 17

Requirement

The Relay Module shall detect communication timeout with CORE

Requirement 18

Requirement

The Relay Module shall transition to its outputs to predefined safe state when timeout with CORE

Requirement 19

Requirement

The Relay Module shall transition to defined safe state when in failure state

Requirement 20

Requirement

The Relay Module shall prevent unintended activation when in failure state

Requirement 21

Requirement

The Relay Module shall provide sufficient fault information to CORE when in failure state

Requirement 22

Requirement

The Relay Module shall execute commands deterministically

Requirement 23

Requirement

The Relay Module shall manage its eight physical driver outputs

Requirement 24

Requirement

The Relay Module shall perform local hardware diagnostics

Requirement 25

Requirement

The Relay Module shall report its status to CORE

Requirement 26

Requirement

The Relay Module shall report its faults to CORE

Requirement 27

Requirement

The Relay Module shall report the state of all eight outputs

Requirement 28

Requirement

The Relay Module shall report the module health

Requirement 29

Requirement

The Relay Module shall report individual channel faults

Requirement 30

Requirement

The Relay Module shall report communication status

Requirement 31

Requirement

The Relay Module shall report electrical identification

Requirement 32

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The Relay Module shall report thermal identification

Requirement 33

Requirement

The Relay Module shall report diagnostics identification

Requirement 34

Requirement

The Relay Module shall report firmware identification

Requirement 35

Requirement

The Relay Module shall report hardware identification

Requirement 36

Requirement

The Relay Module shall report any condition preventing reliable output control

Requirement 37

Requirement

The Relay Module shall reliably energize its assigned relay coil across the full specified supply

Requirement 38

Requirement

The Relay Module shall independently energize its assigned relay coil across the full specified supply

Requirement 39

Requirement

The Relay Module shall energize its assigned relay coil across the full specified temperature

Requirement 40

Requirement

The Relay Module shall energize its assigned relay coil across the full specified tolerance range

Requirement 41

Requirement

The Relay Module shall energize its assigned relay coil with adequate voltage

Requirement 42

Requirement

The Relay Module shall energize its assigned relay coil with adequate voltage margin

Requirement 43

Requirement

The Relay Module shall energize its assigned relay coil with adequate current margin

Requirement 44

Requirement

The Relay Module shall energize its assigned relay coil with continuous-duty

Requirement 45

Requirement

The Relay Module shall energize its assigned relay coil with continuous-duty capability

Requirement 46

Requirement

The Relay Module shall energize its assigned relay coil with short-circuit protection

Requirement 47

Requirement

The Relay Module shall energize its assigned relay coil with overload protection

Requirement 48

Requirement

The Relay Module shall energize its assigned relay coil with no unintended activation during reset conditions

Requirement 49

Requirement

The Relay Module shall energize its assigned relay coil with no unintended activation during fault conditions

Requirement 50

Requirement

The Relay Module shall operate with coils that have a defined nominal voltage

Requirement 51

Requirement

The Relay Module shall operate with coils that have allowable voltage range

Requirement 52

Requirement

The Relay Module shall operate with coils that have nominal current

Requirement 53

Requirement

The Relay Module shall operate with coils that have maximum current

Requirement 54

Requirement

The Relay Module shall operate with coils that have defined resistance

Requirement 55

Requirement

The Relay Module shall operate with coils that have defined inductance

Requirement 56

Requirement

The Relay Module shall operate with coils that have defined inrush behavior

Requirement 57

Requirement

The Relay Module shall operate with coils that have defined continuous-duty rating

Requirement 58

Requirement

The Relay Module shall operate with coils that have operate characteristics that are fully compatible with its corresponding driver

Requirement 59

Requirement

The Relay Module shall operate with coils that have release characteristics that are fully compatible with its corresponding driver

Requirement 60

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range

Requirement 61

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including component tolerances

Requirement 62

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including steady-state variation

Requirement 63

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including startup conditions

Requirement 64

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including brownouts

Requirement 65

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including overvoltage

Requirement 66

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including ripple

Requirement 67

Requirement

The Relay Module shall operate correctly over the complete specified nominal supply range, including transient disturbances from cabinet

Requirement 68

Requirement

The Relay Module shall have drivers with appropriately sized suppressions

Requirement 69

Requirement

The Relay Module shall have drivers with appropriately sized suppressions, such as a flyback diode

Requirement 70

Requirement

The Relay Module shall have drivers with appropriately sized suppressions, such as a TVS/clamp network

Requirement 71

Requirement

The Relay Module shall limit the coil-generated transient to a safe level without exceeding driver ratings

Requirement 72

Requirement

The Relay Module shall limit the coil-generated transient to a safe level without causing unacceptable relay release-time delays

Requirement 73

Requirement

The Relay Module shall transition all relay outputs to defined safe state during out-of-range brownout

Requirement 74

Requirement

The Relay Module shall transition all relay outputs to defined safe state during power loss

Requirement 75

Requirement

The Relay Module shall initialize deterministically with outputs disabled until valid control from the CORE is established after power restoration